AP PRECALCULUS

Unit 1 • Lesson 1.9 practice

Lesson 1.9 • Rational Functions and Vertical Asymptotes

4 multiple-choice questions and 1 free-response question

Question 1

Which line is a vertical asymptote of $r(x) = \dfrac{x+2}{x^2-x-6}$?

(A) $x = -2$

(B) $x = 3$

(C) $x = -2$ and $x = 3$

(D) $x = 6$

Question 2

$\lim\limits_{x \to 1^+} \dfrac{2x}{x-1}$ is

(A) $+\infty$

(B) $-\infty$

(C) 2

(D) 0

Question 3

$\lim\limits_{x \to -2^-} \dfrac{1}{(x+2)^2}$ is

(A) $-\infty$

(B) 0

(C) $\dfrac{1}{4}$

(D) $+\infty$

Question 4

Which function has a vertical asymptote at $x = 4$?

(A) $\dfrac{x-4}{x+4}$

(B) $\dfrac{x^2-16}{x-4}$

(C) $\dfrac{x}{(x-4)^2}$

(D) $\dfrac{(x-4)^2}{x}$

Question 5 (Free response)

Let $r(x) = \dfrac{x-1}{(x+2)(x-3)}$.

(a) Find the vertical asymptotes of the graph of r.

(b) For each vertical asymptote $x = a$, find $\lim\limits_{x \to a^-} r(x)$ and $\lim\limits_{x \to a^+} r(x)$. Show how you determine the signs.